DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

ANO 2025/2026

REDES DE COMUNICAÇÕES II

LABORATORY GUIDE NO. 2:

VIRTUAL LANS + SPANNING TREE

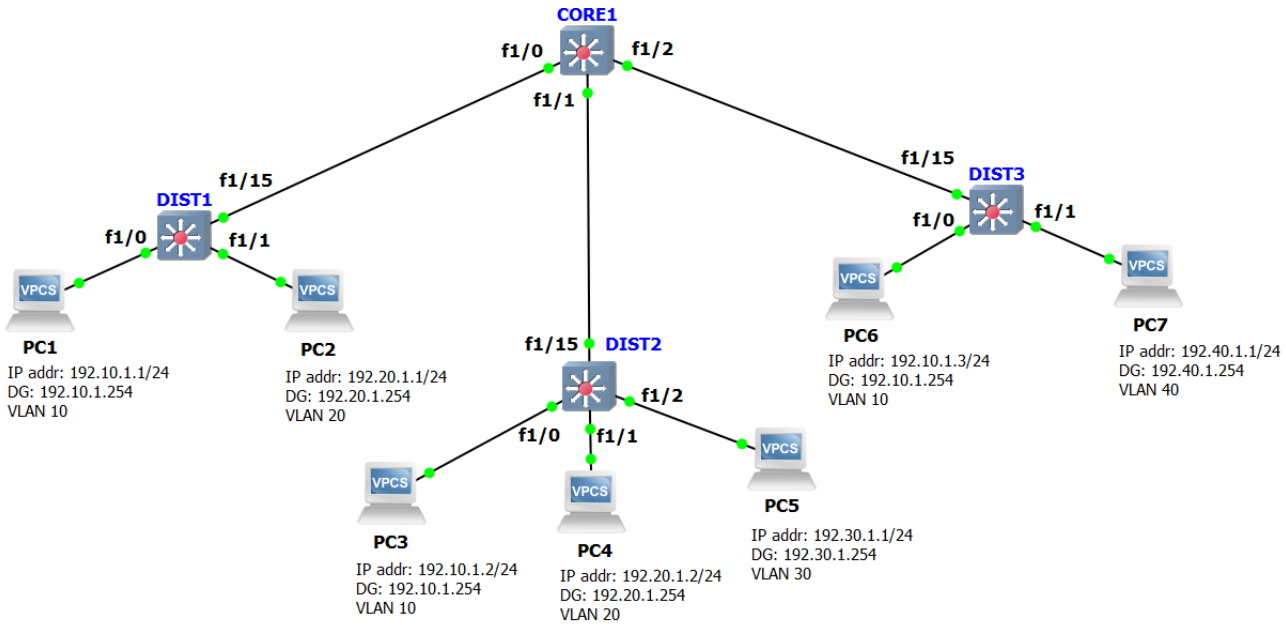
In this Laboratory Guide, all switches should be implemented as L3 Switches with the IOS image 12.4(21) of routers 3725 (provided in the elearning page of RC II) and with two network adapters:

- GT96100-FE, providing 2 FastEthernet routing interfaces: `f0/0` and `f0/1`
- NM-16ESW, providing 16 FastEthernet switching interfaces: `f1/0`, `f1/1`, …, `f1/15`

NOTE: Choose option "This is an EtherSwitch router" when defining the template of the L3 Switches.

1. Virtual LANs

Consider the following network setup composed of 1 core switch (CORE1), 3 distribution switches (DIST1, DIST2 and DIST3) and 7 PCs. Create a GNS3 template with all equipment and all links. Run the template and configure the IP and Default Gateway addresses of all PCs.



Configuration of IP address and Default Gateway in PC1:

```
PC1> ip 192.10.1.1/24 192.10.1.254
PC1> save
```

Check the resulting configuration:

```
PC1> show ip
```

1.a. By default, all switches are working as L2 Switches since all used interfaces are switching interfaces. Run the command `show vlan-switch` on each switch to check that all interfaces belong to VLAN 1 (the default VLAN).

Configure all switches to support all existing VLANs, configure the interfaces of the switches connected to the PCs in the VLANs specified in the figure and configure all links between switches as trunk links with 802.1q encapsulation supporting all VLANs.

Configuration of VLANs, access and trunk ports in switch DIST1:

```
DIST1# vlan database
DIST1(vlan)# vlan 10
DIST1(vlan)# vlan 20
DIST1(vlan)# vlan 30
DIST1(vlan)# vlan 40
DIST1(vlan)# exit
DIST1# configure terminal
```

```
DIST1(config)# interface f1/0
DIST1(config-if)# switchport mode access
DIST1(config-if)# switchport access vlan 10
DIST1(config-if)# interface f1/1
DIST1(config-if)# switchport mode access
DIST1(config-if)# switchport access vlan 20
DIST1(config-if)# interface f1/15
DIST1(config-if)# switchport mode trunk
DIST1(config-if)# end
DIST1# write
```

Check the resulting configuration:

```
DIST1# show configuration
DIST1# show vlan-switch
```

1.b. Start two Wireshark captures: one in link CORE1-DIST1 and another in link CORE1-DIST3. Execute a ping command from each PC to each IP address as specified next. Register the result of each ping and the ICMP and ARP packets observed on each of the two captures. In the analysis of the captured packets, register also to which VLAN each packet belongs. Justify the results.

From:	To:	Result of Ping	Packet in CORE1-DIST1	Packets in CORE1-DIST3
PC1	192.10.1.2			
PC1	192.10.1.3			
PC1	192.10.1.10			
PC1	192.20.1.2			
PC2	192.20.1.2			
PC2	192.30.1.1			

IMPORTANT: Save the current GNS3 template to be used as the initial setup of each of the following sections.

2. Distributed Routing between VLANs

Open the GNS3 template saved in Section 1 and save it as a new template. In the new template, the aim is to configure the IP routing between VLANs with a distributed approach as follows:

(a) VLAN 10 and VLAN 20 are end-to-end VLANs (i.e., they exist in different distribution switches). Therefore, CORE1 should route the packets from/to the IP networks associated with these VLANs.
   - In addition, since VLAN 20 exists only in distribution switches DIST1 and DIST2, VLAN 20 should be removed from the trunk CORE-DIST3.

(b) VLAN 30 is a local VLAN in DIST2. Therefore, DIST2 should route the packets from/to the IP network associated with VLAN 30 and VLAN 30 should be removed from all trunk links.

(c) VLAN 40 is a local VLAN in DIST3. Therefore, DIST3 should route the packets from/to the IP network associated with VLAN 40 and VLAN 40 should be removed from all trunk links

(d) Since CORE1, DIST2 and DIST3 need to route IP packets between VLANs, an interconnection VLAN (with a new associated IP network address) should be configured between them to allow IP routing between all existing VLANs.

Run the new template and configure it following the previously described distributed approach. In step (d), consider VLAN 50 as the interconnection VLAN with the associated IP network 192.1.1.0/24. In this new VLAN 50, assign the IP addresses 192.1.1.1 to CORE1, 192.1.1.2 to DIST2 and 192.1.1.3 to DIST3.

Addition of VLAN 50 and elimination of VLANs in the trunk links of CORE1:

```
CORE1# vlan database
CORE1(vlan)# vlan 50
CORE1(vlan)# exit
CORE1# configure terminal
CORE1(config)# interface f1/0
CORE1(config-if)# switchport trunk allowed vlan remove 30,40
CORE1(config-if)# interface f1/1
CORE1(config-if)# switchport trunk allowed vlan remove 30,40
CORE1(config-if)# interface f1/2
CORE1(config-if)# switchport trunk allowed vlan remove 20,30,40
CORE1(config-if)# end
CORE1# write
CORE1# show configuration    (to check the resulting configuration)
```

Routing configuration of CORE1:

```
CORE1# configure terminal
CORE1(config)# ip routing                    (to activate IP routing)
CORE1(config)# interface vlan10
CORE1(config-if)# ip address 192.10.1.254 255.255.255.0
CORE1(config-if)# no shutdown
CORE1(config-if)# no autostate
CORE1(config-if)# interface vlan20
CORE1(config-if)# ip address 192.20.1.254 255.255.255.0
CORE1(config-if)# no shutdown
CORE1(config-if)# no autostate
CORE1(config-if)# interface vlan50
CORE1(config-if)# ip address 192.1.1.1 255.255.255.0
CORE1(config-if)# no shutdown
CORE1(config-if)# no autostate
CORE1(config-if)# end
CORE1# write
```

Check the resulting configuration:

```
CORE1# show ip interface brief
```

Check the resulting IP routing table:

```
CORE1# show ip route
```

2.a. Check the IP routing tables of CORE1, DIST2 and DIST3 and justify each routing table. Test the connectivity from PC1 to all other existing IP addresses and justify each result.

2.b. What are the connections belonging to the broadcast domain of each existing VLAN? Why?

Now, configure in CORE1, DIST2 and DIST3 the RIPv2 protocol to reach full network IP connectivity.

Configuration of RIPv2 in CORE1:

```
CORE1# configure terminal
CORE1(config)# router rip
CORE1(config-router)# version 2
CORE1(config-router)# network 192.1.1.0
CORE1(config-router)# network 192.10.1.0
CORE1(config-router)# network 192.20.1.0
CORE1(config-router)# end
CORE1# write
```

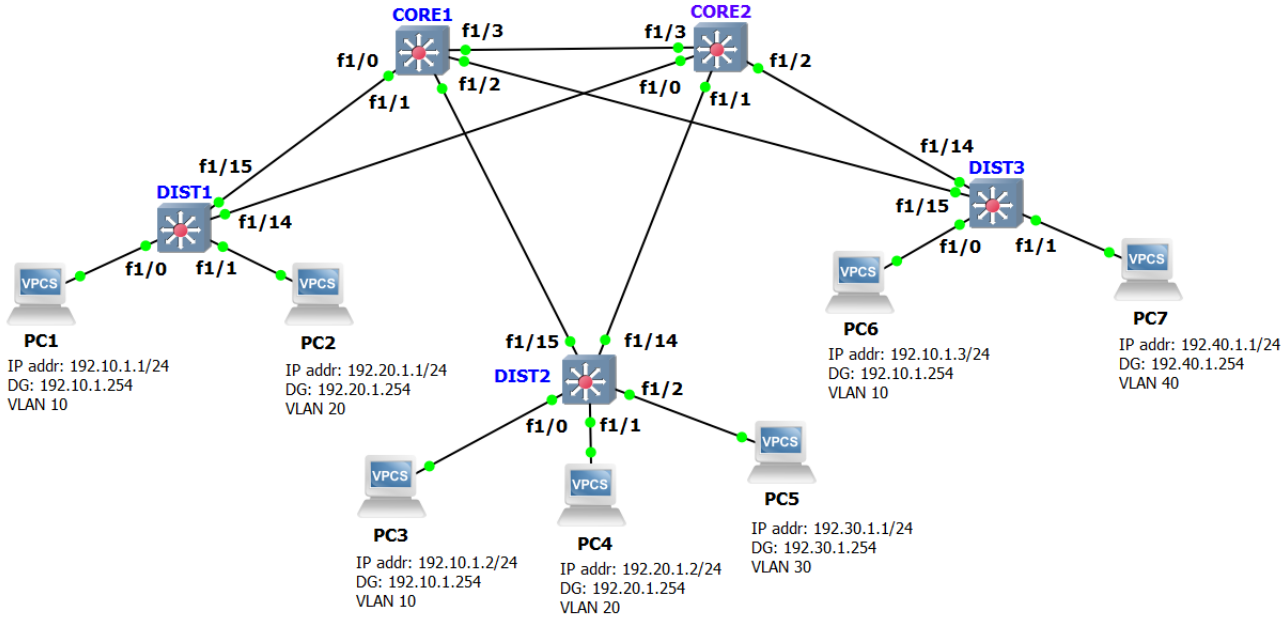
2.c. Check the IP routing tables of CORE1, DIST2 and DIST3 and justify each routing table. Test through ping the connectivity from PC1 to all existing IP addresses (you should conclude that PC1 has connectivity with all addresses and, if not, there are configuration errors that must be identified and corrected).

2.d. Start two Wireshark captures: one in link CORE1-DIST1 and another in link CORE1-DIST3. Execute a ping command from each PC to each IP address as specified next. Register the result of each ping and the ICMP and ARP packets observed on each of the two captures. <u>In the analysis of the captured packets, register also to which VLAN each packet belongs</u>. Justify the results.

From:	To:	Result of Ping	Packet in CORE1-DIST1	Packets in CORE1-DIST3
PC1	192.20.1.1			
PC1	192.20.1.2			
PC1	192.30.1.1			
PC1	192.40.1.1			
PC1	192.1.1.2			
PC7	192.10.1.3			
PC7	192.1.1.3			

3. Redundant LANs with Spanning Tree

Open the GNS3 template saved in Section 1 and save it as a new template (recall that there is no routing configuration between VLANs in this template and all trunk links support VLANs 10, 20, 30 and 40). Before running the new template, add to the network configuration a new core switch (CORE2) with the new connections as shown in the next figure.



Then, run the new template, configure the new inserted links as trunk links and configure the new CORE2 to support all existing VLANs (i.e., 10, 20, 30 and 40).

3.a. Run `show vlan-switch` on each switch to check that all VLANs are correctly configured.

3.b. Run `show spanning-tree bridge` on each switch to identify its *bridge ID* on each VLAN. Based on this information, identify the *root bridge* on each VLAN.

3.c. Run `show spanning-tree brief` on each switch to identify the *port ID* and the *port Cost* of each interface belonging to each VLAN. Check that the spanning tree configurable parameters (*bridge priority* values and *port Cost* values) are the default values in all VLANs.

3.d. The information displayed by the command `show spanning-tree brief` on each switch also enables the identification of the resulting spanning tree for each VLAN. Based on this information:

(i) confirm the ID of the *root bridge* identified in **3.b**,
(ii) identify the state of each interface (*forwarding* or *blocking*),
(iii) identify the ID of the *designated bridge* on each link,
(iv) identify the *root ports* and the *designated ports* of each switch on each VLAN.

Justify the spanning tree of each VLAN based on the parameters observed in **3.b** and **3.c**.

3.e. Start two Wireshark captures: one in link DIST1-CORE1 and another in link DIST1-CORE2. Analyze the STP (spanning tree) packets, justify their content and check that they are periodically sent by the *designated bridge* of each link. Analyze also the PVST packets and confirm that they support the implementation of a spanning tree per existing VLAN. Then, stop the two captures.

3.f. Shutdown interface `f1/3` of CORE2 (simulating a link failure). Run `show spanning-tree brief` on each switch and justify the new spanning tree per existing VLAN. Verify that the new spanning trees recover the logical connectivity between all switches for all VLANs.

Activate interface `f1/3` of CORE2 (i.e., run `no shutdown` on the interface). Different spanning trees can be set up for the different VLANs, as illustrated in the next two experiments.

3.g. Change the *port Cost* of interface `f1/3` in CORE2 to the value 50 only in VLAN 10. Run `show spanning-tree brief` on each switch. Based on the provided information: (i) check that the spanning trees remain the same for VLANs 20, 30 and 40, and (ii) justify the new spanning tree of VLAN 10.

Configuration of the *port Cost* value of CORE2 interface `f1/3` in VLAN 10:

```
CORE2# configure terminal
CORE2(config)# interface f1/3
CORE2(config-if)# spanning-tree vlan 10 cost 50
CORE2(config-if)# end
CORE2# write
```

3.h. Eliminate the previous *port Cost* change (running `no spanning-tree vlan 10 cost 50` on interface `f1/3` of CORE2). Change the *bridge priority* of CORE2 to the value 16384 only in VLAN 10. Run `show spanning-tree brief` on each switch. Based on this information: (i) check that the spanning trees remain the same for VLANs 20, 30 and 40, and (ii) justify the new spanning tree for VLAN 10.

Configuration of the *bridge priority* value of CORE2 in VLAN 10:

```
CORE2# configure terminal
CORE2(config)# spanning-tree vlan 10 priority 16384
CORE2(config)# end
CORE2# write
```

Eliminate the previous *bridge priority* change (running `no spanning-tree vlan 10 priority 16384` on CORE2). Start a never-ending ping command at PC1 to PC6 (at PC1, run `ping 192.10.1.3 -t`). Start two Wireshark captures: one in link DIST1-CORE1 and another in link DIST1-CORE2.

3.i. Register in which link(s) the ICMP Echo Request and Echo Reply packets exchanged between PC1 and PC6 are observed. Justify the observations based on the spanning tree of VLAN 10 (already analyzed in **3.d**).

3.j. Shutdown the interface `f1/0` of CORE1. Register which packets are observed on each link <u>before and after the connectivity between PC1 and PC6 is restored</u>. Justify the observations.

3.k. Activate again the interface `f1/0` of CORE1. Register which packets are captured on each link <u>before and after the connectivity between PC1 and PC6 is restored</u>. Justify the observations.

Stop the two Wireshark captures and the never-ending ping command at PC1 (by inserting `Ctrl+C`).

Configure the routing between all VLANs in both CORE1 and CORE2 selecting appropriately the IP addresses of CORE1 and CORE2 on each VLAN. Then, configure CORE1 and CORE2 with VRRP on each VLAN so that the master default gateway in VLANs 10 and 20 is CORE1 and the master default gateway in VLANs 30 and 40 is CORE2. (**Recall the configuration instructions provided in Laboratory Guide No. 1**)

3.l. Check the IP routing tables of CORE1 and CORE2 and justify each routing table. Test through ping the IP connectivity between all pairs of PCs. If there are some pairs of PCs without connectivity, clear the MAC Adress Tables of both CORE1 and CORE2 (if these actions do not solve the connectivity problem between all PCs, there are configuration errors that must be identified and corrected).

Clear the MAC Address Table of CORE1:

```
CORE1# clear mac-address-table
```

Start a never-ending ping command at PC1 to PC7 (at PC1, run `ping 192.40.1.1 -t`).

3.m. Stop running CORE1 (simulating an equipment failure). Observe that the IP connectivity fails temporarily and recovers after some seconds.

3.n. Start running again CORE1 (simulating an equipment failure recovery). Again, observe that the IP connectivity fails temporarily and recovers after some seconds (it might be needed to clear again the MAC Address Tables of CORE 1 and CORE2 to recover connectivity).

Stop the never-ending ping command at PC1 (by inserting `Ctrl+C`). Explain how the network reacts to recover the IP connectivity in each of the experiments **3.m** and **3.n**.